Posts and Telecommunications Institute of Technology
Campus at Hồ Chí Minh City

DATABASES (INT1313_CLC)

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Lab 2

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In this lesson, you will have hands-on exercises with Database Management Systems (DBMS) and database operations, including programming structures across various programming languages for database interaction and API library development.

A. DBMS Installation and Database Deployment

These tasks were covered during the midterm assignment. Additionally, students are required to configure access controls and user classifications for the databases.
Students may proceed with these lab exercises based on the requirements from the midterm assignment or choose to install one of the following Database Management Systems:

- MS SQL Server 2019 Express: https://www.microsoft.com/en-us/download/details.aspx?id=101064
- MySQL: https://www.oracle.com/mysql/technologies/mysql-enterprise-edition-downloads.html
- PostgreSQL: https://www.postgresql.org/download/
- SQLite: https://sqlite.org/index.html
- Oracle Database Express - Oracle XE: https://www.oracle.com/asean/database/technologies/xe-downloads.html

Then, please do the following tasks:

- Download the sample databases
- Create the database in the installed DBMS
- Import data from the sample into the newly created database
- Execute SQL queries to retrieve information. I will provide the data retrieval requests during Q&A section.

Example: Movies sample database: https://www.databasestar.com/sample-database-movies/

B. Database Programming

Choose one of the following programming languages: PHP, Python, Java, Node.js, or Go (Golang).
Learn to program in your chosen language to:

- Connect to the installed DBMS and database.
- Query and retrieve data from the database and display the results.
- Write data to the database.
- Close the database connection before terminating the program.

C. API Library

Learn how to create an API library that interacts with the database installed and configured in the previous exercise.
Perform tests using the created API library.

Note: API management tools (e.g., Postman, Swagger, or apidog) are highly recommended.

D. Data Sources

Many useful sources provide data from real-world applications. Listed below are some typical sources you can consult and use for your projects:

- https://www.kaggle.com/datasets
- https://huggingface.co/datasets
- https://datasetsearch.research.google.com/

You can find more from other sources.
When using the above data in your projects, remember to clearly state the source of the data along with a detailed description of how the data are used in your project.

Thanh you!

End